## Type MS · Sequence and describe the stages of mitosis

### Answer 1 of 6 · Question 5 part b · 5 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

| number | | | | |
|---|---|---|---|---|
| 5 | (a) (i) | A (DNA);<br><br>B/C/ and D are found in or associated with other structures. | | 1 |
| | (ii) | correct circles of pair; | | 1 |
| | (iii) | only two pairs of homologous chromosomes/ would be 23 in a human cell; | | 1 |
| | (b) | • prophase, metaphase, anaphase and telophase;<br>• (prophase) chromosomes become thicker/visible;<br><br>• (metaphase) chromatids line up at equator;<br>• (anaphase) chromatids separate and migrate to poles;<br>• (telophase) nucleus/nuclear membrane reforms; | Allow condense<br><br>Allow chromosomes throughout<br><br>Allow clear annotated diagrams showing stages | 5 |

Total 8 marks

## Type MS · Sequence and describe the stages of mitosis

### Answer 1 of 6 · Question 5 part b · 5 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Sequence and describe the stages of mitosis

### Answer 2 of 6 · Question 6 part a.iv · 3 marks

2020 NOV · Paper 01 (regional R) · find part a.iv in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

## Type MS · Sequence and describe the stages of mitosis

### Answer 3 of 6 · Question 7 part a.i · 2 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

| number | | | | |
|---|---|---|---|---|
| 7 (a) | (i) | • prophase, metaphase, anaphase and telophase;; | minus 1 for each error | 2 marks |
| | (ii) | • chromosomes lined up;<br>• equator shown/aligned at equator;<br>• one member of each bivalent above, the other below; | | 3 marks |
| (b) | (i) | A (DNA);<br><br>B not part of DNA<br>C not part of DNA<br>D found only in RNA not DNA | | 1 mark |
| | (ii) | length = 20 mm;<br>$\frac{20}{750}$ = 0.026 mm;<br>= 26.0/26.7 µm; | allow 20 -22<br><br>ECF | 3 marks |
| (c) | | • two genetically identical cells;<br>• diploid/2n; | | 2 marks |

Total 11 marks

## Type MS · Sequence and describe the stages of mitosis

### Answer 3 of 6 · Question 7 part a.i · 2 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Sequence and describe the stages of mitosis

### Answer 4 of 6 · Question 7 part a.ii · 3 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

| number | | | | |
|---|---|---|---|---|
| 7 (a) | (i) | • prophase, metaphase, anaphase and telophase;; | minus 1 for each error | 2 marks |
| | (ii) | • chromosomes lined up;<br>• equator shown/aligned at equator;<br>• one member of each bivalent above, the other below; | | 3 marks |
| (b) | (i) | A (DNA);<br><br>B not part of DNA<br>C not part of DNA<br>D found only in RNA not DNA | | 1 mark |
| | (ii) | length = 20 mm;<br>$\frac{20}{750}$ = 0.026 mm;<br>= 26.0/26.7 µm; | allow 20 -22<br><br>ECF | 3 marks |
| (c) | | • two genetically identical cells;<br>• diploid/2n; | | 2 marks |

Total 11 marks

## Type MS · Sequence and describe the stages of mitosis

### Answer 4 of 6 · Question 7 part a.ii · 3 marks

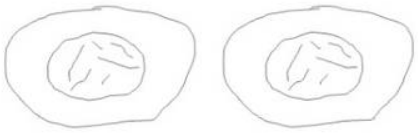
2021 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Sequence and describe the stages of mitosis

### Answer 5 of 6 · Question 2 part b.ii · 1 mark

2022 JUN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 2 (a) (i) | 156 – 72;<br>84; | ECF if candidates use incorrect values from the graph with correct answer from for incorrect values = 1 mark | 2 |
| (ii) | **C** 9 minutes; | | 1 |
| (iii) | In the following order only:<br>• oxygen;<br>• diffusion;<br>• aerobic/cellular;<br>• decreases/reduces/goes down; | | 4 |
| (b) (i) |  <br>must have 6 chromosomes shape not important | One mark for each correct cell drawn | 2 |
| (ii) | **B** metaphase;<br><br>A is incorrect as chromosomes do not align at the equator during anaphase<br>C is incorrect as chromosomes do not align at the equator during prophase<br>D is incorrect as chromosomes do not align at the equator during telophase | | 1 |
| | | | Total 10 |

## Type MS · Sequence and describe the stages of mitosis

### Answer 5 of 6 · Question 2 part b.ii · 1 mark

2022 JUN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

| uestion number | Answer | Notes | Mark s |
|---|---|---|---|

## Type MS · Sequence and describe the stages of mitosis

### Answer 6 of 6 · Question 2 part b.i · 1 mark

2025 JUN · Paper 02 (regional R) · find part b.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 2 (a) | | **C (endoplasmic reticulum)**<br><br>The answer is not<br>A as the cell membrane transports substance into and out of the cell<br>B cytoplasm as this is a base for chemical reactions<br>D as ribosomes provide sites for protein synthesis | | 1 |
| (b) | (i) | Z Y W X; | | 1 |
| | (ii) | growth/repair; | Ignore repair of body cells | 1 |
| (c) | | Any two from<br>• mitosis produces diploid cells/46 chromosomes, meiosis produces haploid cells/23 chromosomes;<br>• mitosis produces 2 (daughter) cells, meiosis produces 4 cells/gametes;<br>• (daughter) cells from mitosis are genetically identical, (daughter) cells from meiosis show variation; | | 2 |
| | | | **Total for Question 2 = 5 marks** | |